

2026

VAD - Project Report

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1 Introduction

As part of the Advanced Data Visualization project, this work aims to tackle the growing complexities of space traffic by developing an interactive visualization tool for satellite orbits. The primary motivation behind this project is the rapid increase in objects orbiting Earth, which raises significant concerns regarding space safety and asset management.

The specific problem being addressed is the difficulty of intuitively understanding and predicting risks in an increasingly congested orbital environment. By transforming complex orbital data into a communicative and analytical visualization, this project seeks to make these risks immediately apparent to end users.

Our target users—including satellite operators, aerospace engineers, and researchers—need to extract insights regarding satellite density, debris localization, asset operational status, atmospheric reentries, and collision risks to safely monitor space assets and plan trajectories.

2 Related Work

The development of this interactive orbital visualization is informed by an analysis of existing platforms dedicated to satellite tracking and Space Domain Awareness (SDA). These tools serve as a foundational benchmark for our design premises, allowing for a comparative study of user interface (UI) responsiveness, data filtering capabilities, and spatial representation strategies. By examining the strengths and limitations of current solutions, we aim to implement a visualization that balances high-fidelity 3D rendering with intuitive analytical features.

Several platforms provide live 3D representations of the Earth and orbital objects. These systems typically feature data filtering and provide detailed orbital parameters upon user interaction. Platforms such as Satellite Tracker 3D [1] and Satellite Map [3] offer extensive filtering based on mission types (e.g., Weather, Navigation) and provide a more responsive UI. In contrast, LeoLabs [2] prioritizes high-fidelity visualization by rendering distinct 3D models for Satellites, Rocket Bodies, and Debris. While this level of detail results in a more resource-intensive interface, it allows for the visualization of radar beams and ground instruments. These sensors refer to phased-array radars capable of detecting and tracking small orbital fragments in real-time. However, LeoLabs offers fewer categorical filtering options compared to its counterparts.

In terms of operational safety and risk mitigation, Neuraspace Space Traffic Management Platform [4] represents a significant advancement in operational safety, being the first spaceops platform

to specifically address automated collision avoidance. While other tools focus on historical data or visual tracking, Neuraspace prioritizes real-time Space Traffic Management (STM) through AI-driven risk assessment. This focus on operational intelligence and collision mitigation is highly relevant to our analytical goals, as it highlights the importance of transitioning from static monitoring to proactive orbital safety management.

The interface design was developed through a comparative analysis of diverse dashboard paradigms. For the Orbit Page, the layout is a result of design synthesis, merging various visualization strategies into a cohesive multi-view environment that balances 3D spatial awareness with categorical data ([8] [9] [10]). In contrast, the Launch Site Page was more directly influenced by a specific dashboard [11]. This reference served as a structural benchmark for organizing real-time KPI blocks and hierarchical ranking lists, ensuring a clean and efficient information architecture for operational data.

3 Design requirements

The main goal is to provide visual analytics of orbital objects and launch sites. To ensure that our visualization effectively addresses the complexities of space traffic management, we have defined the following analytical questions, user tasks, and system design requirements:

3.1 Analytical Questions

- What is the percentage of each object type in orbit?
- Which operational satellites currently in orbit belong to a specific company/country, and what are their current orbital trajectories?
- How has the number of satellites in orbit changed over time?
- How are satellites distributed by purpose (e.g., communication, navigation, scientific)?
- How does satellite density vary across different orbital regions (e.g., LEO, MEO, GEO)?
- What proportion of satellites in orbit are active versus inactive, based on the displayed data?
- What is the minimum distance between two specific orbiting bodies within a given timeframe?
- How are global launch infrastructures geographically distributed, and which specific sites exhibit the highest operational density in terms of total deployments?
- How has the usage of specific launch sites changed and grown over the decades?

3.2 System Requirements

The functional and technical requirements mandate that the system must: support filtering by object status (active, inactive, decayed) and type, enable spatial exploration across orbital bands (LEO, MEO, GEO), feature cross-filtering interactions (e.g., selecting a country filters the 3D map), and display details-on-demand (metadata and proximity/conjunction distances) via interactive hover or click events.

4 Data

The new datasets created for this project was created from three different sources: Celestrak [5], Space-Track [6] and Jonathan’s Space Report [7]. Each of them gave us different information from

the orbital objects. The code files used to do this merge were: `celes_create_dataset.ipynb` and `datasets_merged.ipynb`.

Celestrak was primarily used for its supplemental information data such as Weather, Earth Resources, Communications, Navigation, Scientific and Miscellaneous Satellites. All these data were scattered through different files that had to be joined in a single csv file.

The process of joining the datasets started with the analysis, cleansing, and typo correction of Jonhathan’s Report. Followed by the verification of missed objects between this dataset and Celestrak’s using the NORAD CAT IDs (Unique Satellite Catalog Number). During this process, the datasets’ different nomenclatures were taken into account. Information about space stations and objects in analyses was added. The unnecessary columns for the project were dropped.

Finally, for the positional data, Space-Track was used to retrieve TLE information. TLE is a two-line format that describes the position of orbital objects at a specific time. Space-Track has a comprehensive set of Orbital Mean Elements (OMM) that are often simplified or omitted in standard CelesTrak CSV exports. Specifically, it provides the Right Ascension of the Ascending Node (RAAN), the Argument of Perigee, and the Mean Anomaly, that are not present on the CelesTrak’s dataset. (Figure 1)

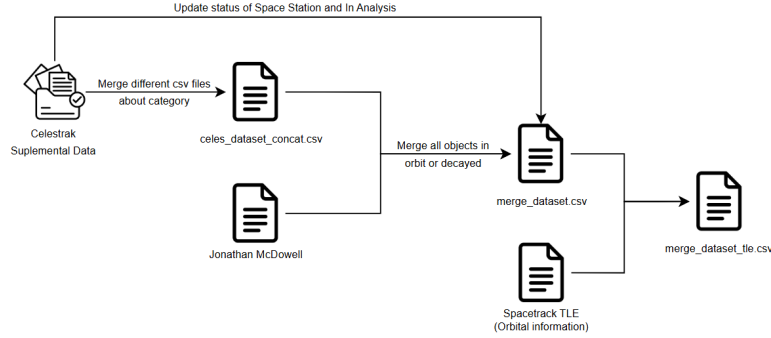


Figure 1: Merged Dataset pipeline.

Based on the unique launch sites identified in the final dataset, we mapped their respective geographical coordinates and host countries. This information was compiled into a dedicated CSV file to enable the spatiotemporal visualization of launch activity and its evolution throughout the years (`launch_site_gps.csv`).

4.1 Columns

The structured merged dataset comprises: numerical identifiers and dimensions (e.g., *NORAD_CAT_ID*, *MASS*, *LENGTH*); orbital elements (*INCLINATION*, *ECCENTRICITY*, *APOGEE*, *PERIGEE*); ISO 8601 timestamps (*LAUNCH_DATE*, *DECAY_DATE*, *EPOCH*); and binary flags (0 or 1) mapping satellite missions and constellations (e.g., *STARLINK*, *COMMUNICATION*).

The auxiliary launch infrastructure dataset organizes spatial and operational attributes, comprising unique facility identifiers (*LAUNCH_SITE*) alongside descriptive localization fields (*LOCATION_NAME*, *COUNTRY*). To support geographical mapping and high-density statistical ranking, this structure pairs precise geospatial coordinates (*LATITUDE*, *LONGITUDE*) with aggregated historical mission deployment volumes (*count*).

5 Exploratory Data Analysis

This section summarizes the key insights obtained from the exploratory data analysis (EDA) of orbital objects and launch site data. Each visualization supports the analytical questions and

5.3 Spatial Distribution and Orbital Density

Analyzing the spatial distribution addresses how satellite density varies across different regions, a critical factor for space safety.

- **Macro-Level Architecture:** The 3D visualization (Figure 4) provides a clear perspective of the orbital architecture: a highly dense, spherical cloud of objects enveloping the Earth (LEO), surrounded by a distinct, sparse outer ring representing the Geostationary belt (GEO).
- **Orbital Regimes:** The vast majority of tracked objects (86.2%) reside in Low Earth Orbit (Figure 16), confirming it as the most high-risk operational domain.
- **Altitude and Inclination:** The violin and scatter plots (Figures 5a to 27) demonstrate distinct clustering based on mission profiles. LEO exhibits dense, widely distributed bands at varying altitudes and inclinations, whereas MEO and GEO show much tighter operational bands. Specifically, the GEO distribution (Figure 21) reveals a strict, narrow altitude convergence (35,786 km) dedicated almost exclusively to Communication, Weather, and Earth Resources satellites.

These spatial insights relate to the **Explore Orbital Distribution** requirement, ensuring users can navigate between the chaotic, multi-purpose LEO environment and the highly structured GEO belt.

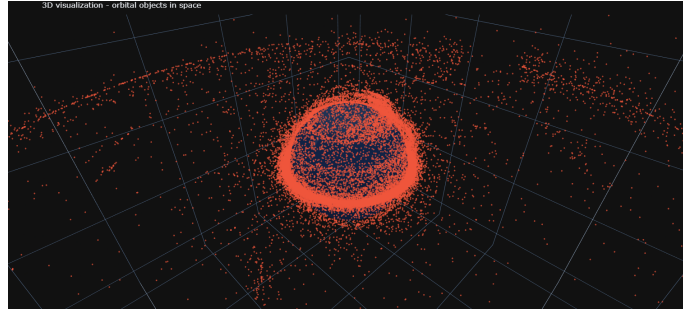


Figure 4: 3D Visualization of Orbital Objects (Plotly WebGL)

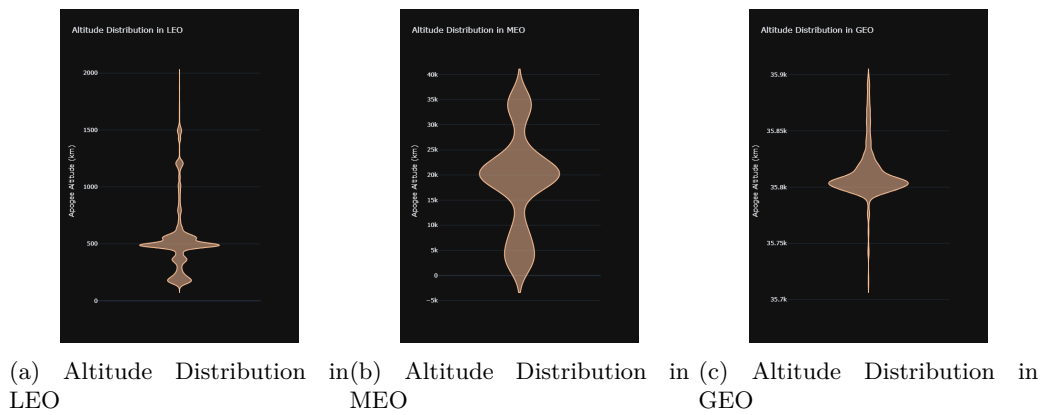


Figure 5: Altitude distribution of objects in the Orbital Regions

5.4 Satellite Categories influence in Mass and Lifespan of reentered objects

Satellites vary significantly depending on their purpose. The EDA highlights the distribution of median mass by mission category (Figure 29), revealing sharp contrasts between heavy GEO

communication satellites and lighter, mass-produced LEO constellations. Furthermore, the orbital lifetime distribution of already decayed objects indicates an average lifespan of roughly 2.92 years, though with significant variance 30.

5.5 Risk Assessment and Proximity Analysis

Finally, to address the need for collision risk prediction, we analyzed minimum distance metrics within specific timeframes. Figure 32 (Top 10 Nearest Space Objects to the ISS over 7 days) serves as a proof-of-concept for risk of collision assessment. It demonstrates the feasibility of calculating and displaying close approaches between orbiting bodies. This directly shows the **Assess Asset Proximity** functional requirement, ensuring that satellite operators can interact with the final visualization tool to show the proximity of their satellites to other operational trajectories.

5.6 Global Launch Infrastructure and Historical Trends

To understand the source of this congestion, we analyzed the geographic distribution and operational history of global launch infrastructures.

- **Geographic Dominance:** The data reveals a highly concentrated launch landscape. The United States, Russia, Kazakhstan, and China dominate historical launch volumes (Figure 6b), which is visually corroborated by the global map of launch sites (Figure 6a).
- **Site Raking:** Top launch sites, namely Cape Canaveral (USA), Plesetsk Cosmodrome (Russia), and Baikonur Cosmodrome (Kazakhstan), account for the vast majority of historical deployments (Figure 33).
- **Decadal Shifts:** Analyzing specific sites over time reveals shifting geopolitical and commercial dynamics. For example, while Russia’s PLMSC (Figure 35) shows strong historical dominance with recent fluctuations, China’s TAISC (Figure 36) displays aggressive, sustained growth since 2019. Most notably, the USA’s AFETR (Figure 34) illustrates an unprecedented, exponential spike in recent years, heavily correlated with the rise of commercial mega-constellations. (For the purpose of this report, only three representative launch sites are analyzed.)

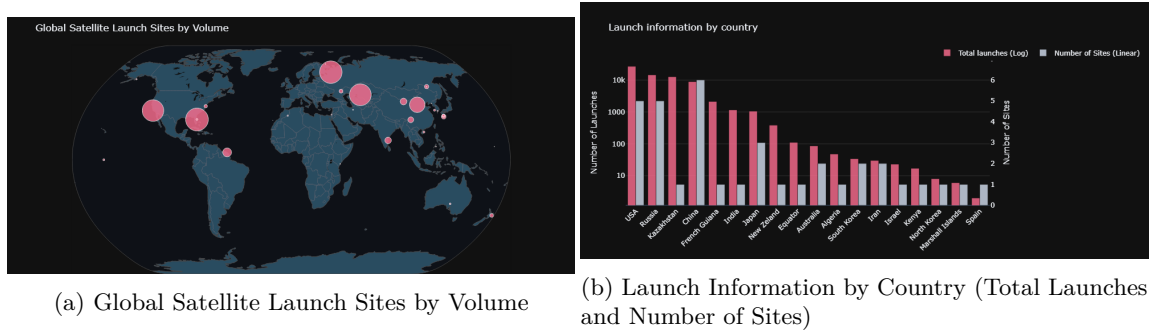


Figure 6: Launch sites EDA plots.

6 Design

The dashboard architecture is organized into two distinct functional views: the Orbital Visualization page and the Launch Sites Analysis page. Navigation between these views is managed through a global header containing persistent navigation buttons. To optimize screen real estate and avoid visual clutter, a vertical overflow strategy was implemented, allowing users to explore the data through scrolling while maintaining a clean, structured layout as shown in Figure 7.



(a) Orbital page design



(b) Launch Site page design

Figure 7: Design template for dashboard visualization

6.1 Orbit Page

The system is designed around the concept of Multiple Coordinated Views (MCV). By adopting a grouped layout strategy, all visualizations are integrated into a single, synchronized functional unit. This ensures bi-directional interactivity, when a filter or data point is selected in one component, all remaining views are updated in real-time to reflect the current subset.

This page features a coordinated multi-view dashboard composed of a Donut Chart for object type distribution, a Bar Chart ranking ownership by country, a central 3D spatial substrate with a collapsible filter panel, a Violin Plot visualizing altitude density, and a Line Graph tracking historical launch frequency.

Regarding interaction design, the 3D map employs a Details-on-Demand strategy; clicking a specific orbital object triggers a pop-up revealing its precise name and technical orbital parameters. Furthermore, the UI includes parametrization through a collapsible control panel, allowing the user to toggle between a focused map view or a full-data exploration mode.

6.2 Launch Page

The interaction strategy for the Launch Site Page is built upon a filtered exploration model. While the layout maintains a fixed structure to ensure high-density data readability, the user can interact with a selection menu to isolate a specific Launch Site. Upon selection, the temporal evolution charts and performance indicators are dynamically updated to reflect the operational lifecycle and historical mission frequency of the chosen infrastructure.

The view integrates a vertical leaderboard ranking sites by mission volume, high-level KPI blocks detailing recent deployments, a 2D global map with size-encoded launch volumes, a dual-axis chart correlating national infrastructure with operational output, and a temporal line chart to isolate site performance.

Regarding interaction design, the 2D map employs a Details-on-Demand strategy. Hovering over a specific launch site triggers a tool-tip or pop-up revealing the distribution of launches by country for that location.

7 Implementation

The final platform’s interface architecture is structured into two primary functional pages, dedicated to satellite monitoring and global launch sites, and integrated into scrollable dashboard environment. Users navigate the spatial data through an interactive 3D globe substrate that supports intuitive rotation and zoom controls for detailed orbital inspection. To manage and dissect data complexity, the interface deploys categorical checkbox filtering panels, allowing users to dynamically isolate specific satellite mission profiles and launch facility networks based on their active analytical needs (Figures 8 & 9).

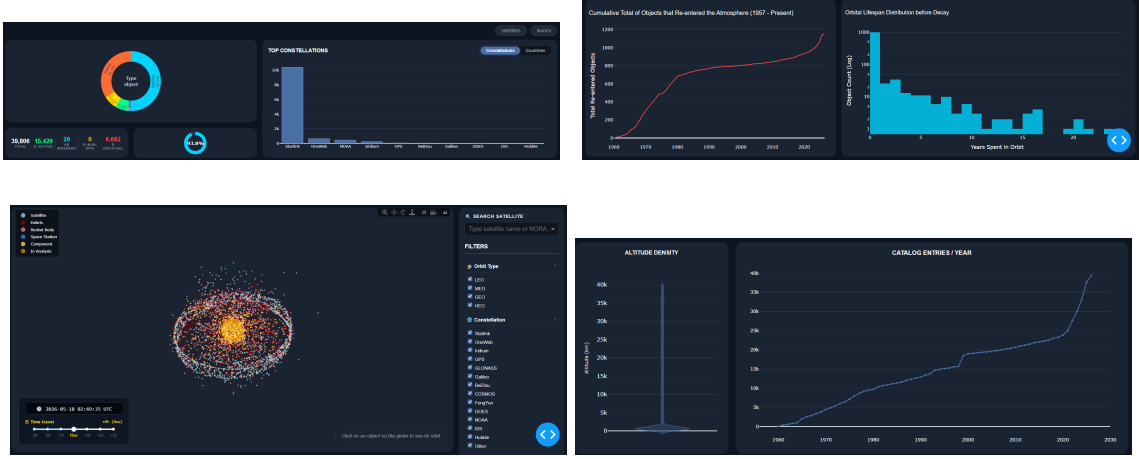


Figure 8: Full Satellite Dashboard Overview.



Figure 9: Full Launch Dashboard Overview.

The true analytical depth of the application is driven by multi-view coordination and rich interaction mechanisms that seamlessly connect spatial rendering with statistical plots. Any modification within the global filter panels synchronously updates all remaining charts on the page to maintain cross-filtering consistency. On the Launch view, selecting a data point within the comparative bar plot dynamically updates the 2D global map to pinpoint and isolate the corresponding geographic site. On the Satellite view, clicking any individual space object triggers a details-on-demand overlay that instantly displays its exact orbital trajectory around the Earth. Finally, to support predictive safety analysis and historical tracking, a spatiotemporal "time travel" mechanism enables users to simulate and visualize exact satellite positions within a dynamic range of three days into both the past and the future.

8 Evaluation

For evaluating our final visualization, we gathered feedback and opinions from classmates, as well as discussion between the two authors of this project. Analytical questions were often consulted to ensure that the project direction aligned with them. To validate 3D orbits and object positions in the globe, Satellite Tracker 3D [1] was frequently visited and compared.

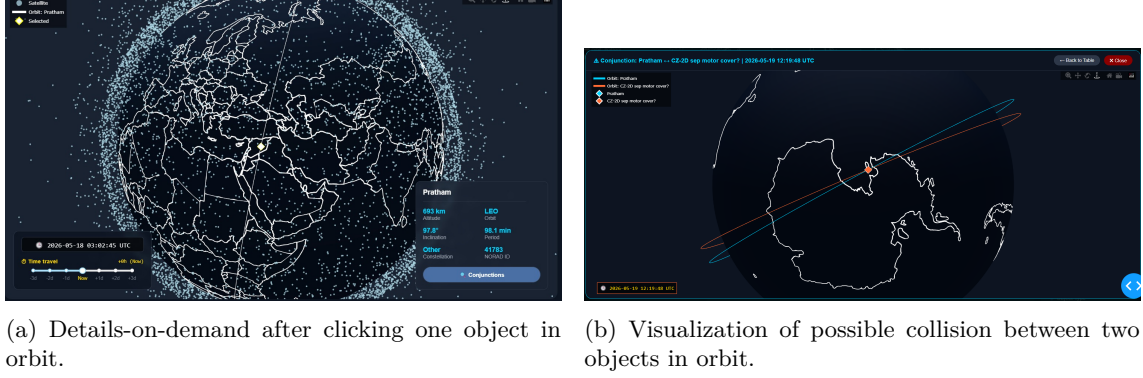


Figure 10: Interactions with the dashboard.

One of the main strengths of the implemented system is the coordination between the diverse visualizations, which allows users to retrieve multi-faceted information about our merged dataset while interacting with the data/plots. An important positive output of the visualization is the capability to successfully address all the questions stated in Section 3.

Areas for future improvement primarily involve the smoothness of the objects live movement, with the application of interpolation between calculated points. Additionally, optimizing the globe’s refresh mechanism is necessary to prevent data updates from disrupting the 3D view while the user is actively interacting with the environment.

9 Critical Reflection

The project team is highly satisfied with the final platform, which successfully unifies 3D space monitoring with categorical analytics in a single environment. The coordinated multi-view architecture proved exceptionally effective, providing clear visual answers to all analytical questions defined in Section 3.

The primary value of this visualization lies in revealing spatial dynamics that are impossible to comprehend through abstract, static TLE tables. The interactive 3D rendering enabled an immediate architectural contrast between the chaotic spherical cloud of Low Earth Orbit (LEO) and the highly structured linear ring of Geostationary Orbit (GEO). Furthermore, dynamic cross-filtering facilitated the immediate analysis of close-approach vectors and imminent collision risks (such as the ISS conjunction), delivering a level of three-dimensional situational awareness completely inaccessible through traditional text-based methods.

Under fewer constraints, future iterations would prioritize position interpolation algorithms for continuous satellite movement and optimize camera state-preservation during backend refreshes to prevent user interaction disruptions. Finally, a high-precision localized zoom feature would significantly enhance the dissection of tight collision trajectories.

References

- [1] Satellite Tracker 3D, *Satellite Tracker 3D – NORAD ID 58258*. Available at: <https://satellitetracker3d.com/track?norad-id=58258>
- [2] LeoLabs, *LeoLabs – Space Domain Awareness Services*. Available at: <https://leolabs.space/>
- [3] SatelliteMap.space, *SatelliteMap.space – Satellite Visualization Platform*. Available at: <https://satellitemap.space/>

- [4] Neuraspace, *Neuraspace Projects*. Available at: <https://www.neuraspace.com/projects>
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- [7] McDowell, Jonathan C., 2020. General Catalog of Artificial Space Objects, Release 1.8.0 , <https://planet4589.org/space/gcat>
- [8] Geckoboard, *Geckoboard Data Dashboard Software*, [Online Platform]. Available: <https://www.geckoboard.com/about/> [Accessed: 04-Apr-2026].
- [9] ZoomCharts, *Microsoft Power BI Custom Visuals and Dashboard Examples*, [Online Platform]. Available: <https://zoomcharts.com/en/microsoft-power-bi-custom-visuals/dashboard-and-report-examples/> [Accessed: 04-Apr-2026].
- [10] Dribbble, *Dashboard Design Inspiration and Portfolio*, [Online Database]. Available: <https://dribbble.com/search/dashboard> [Accessed: 04-Apr-2026].
- [11] Geckoboard, *Daily Pulse Company Dashboard Example*, [Online Image]. Available: <https://www.geckoboard.com/uploads/daily-pulse-company-dashboard-example.png> [Accessed: 04-Apr-2026].

A Exploratory Data Analysis (EDA) Figures - Part 2

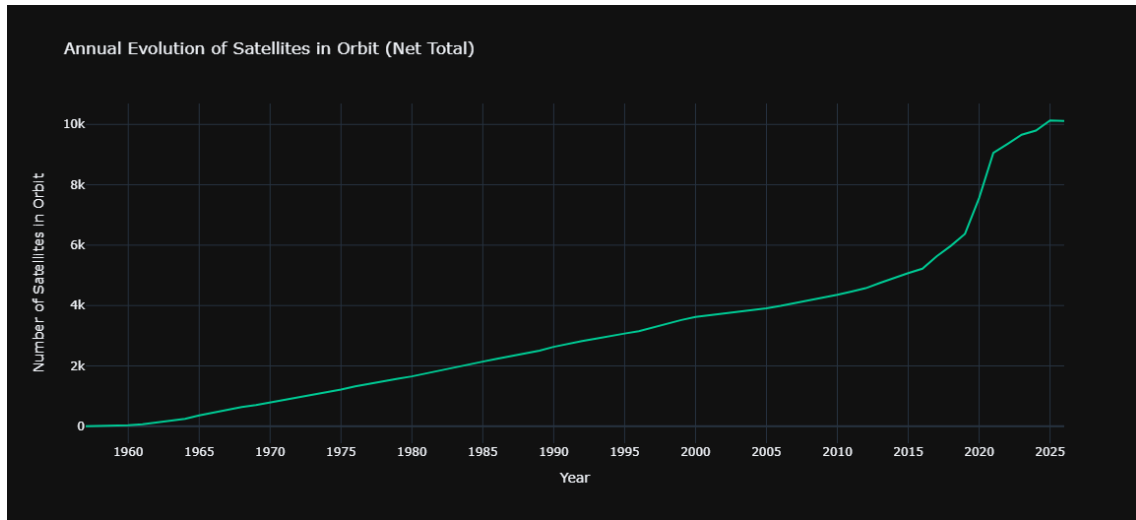


Figure 11: Annual Evolution of Satellites in Orbit

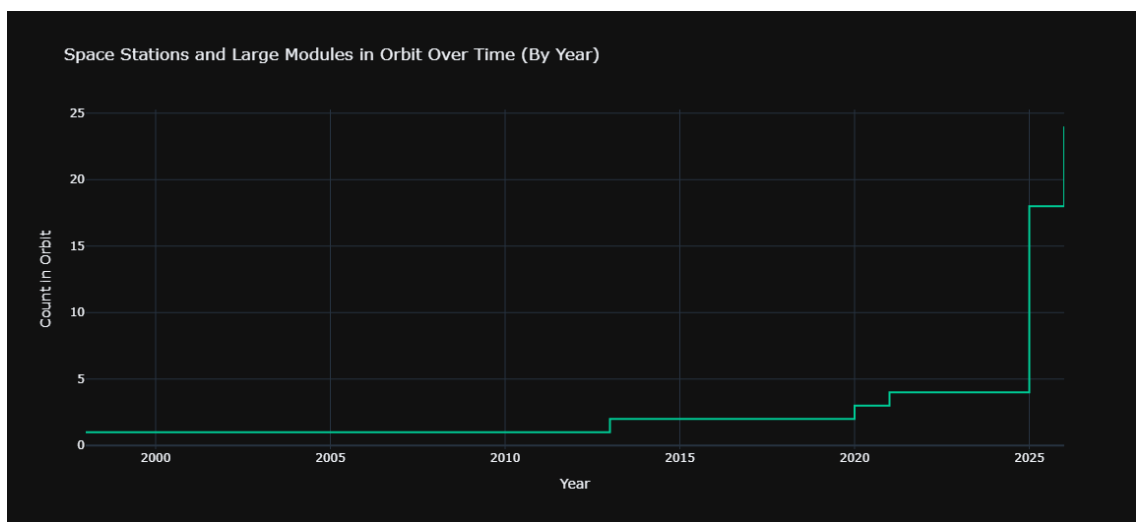


Figure 12: Space Stations and Large Modules in Orbit Over Time

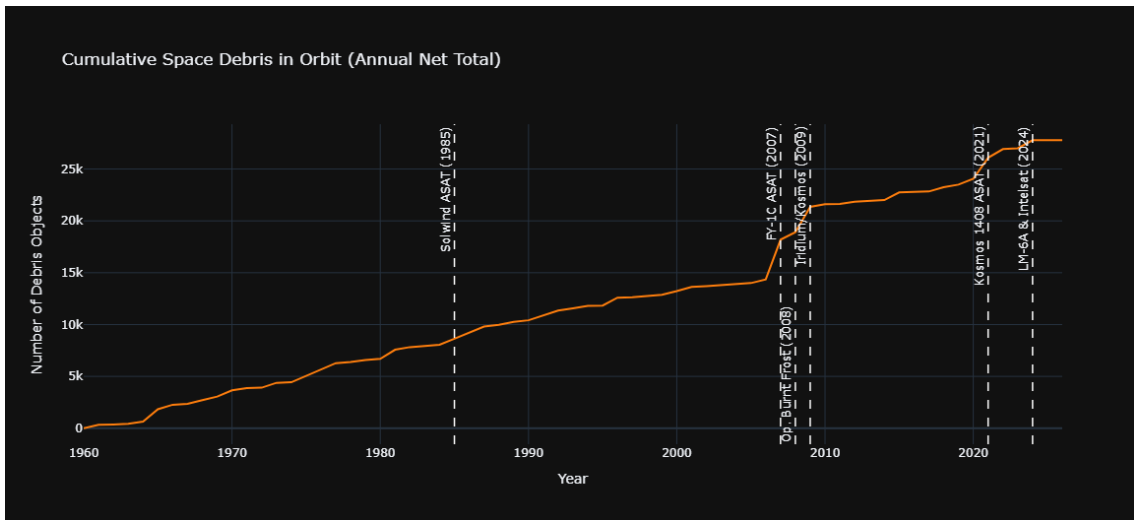


Figure 13: Cumulative Space Debris in Orbit

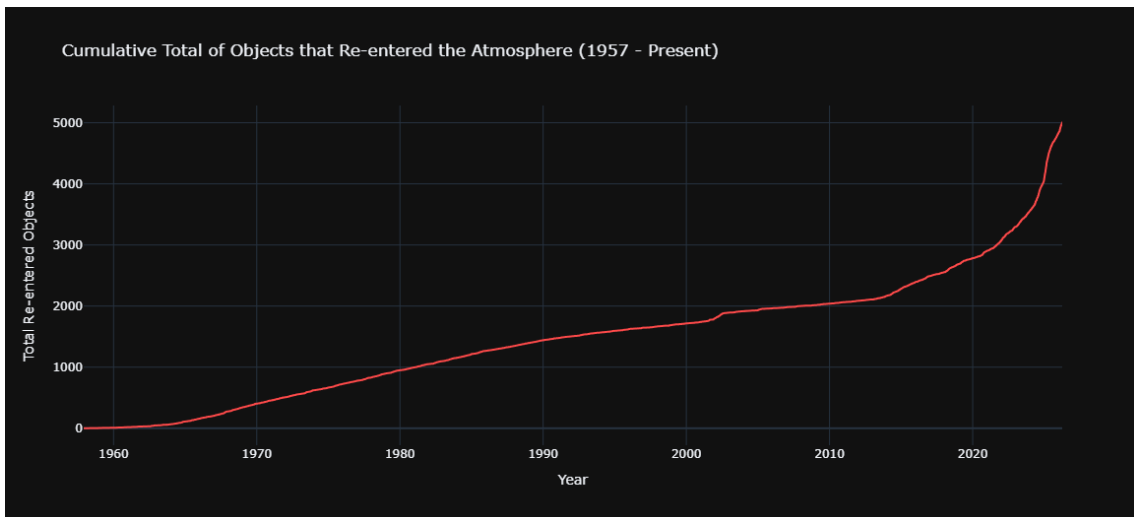


Figure 14: Cumulative Total of Objects that Re-entered the Atmosphere

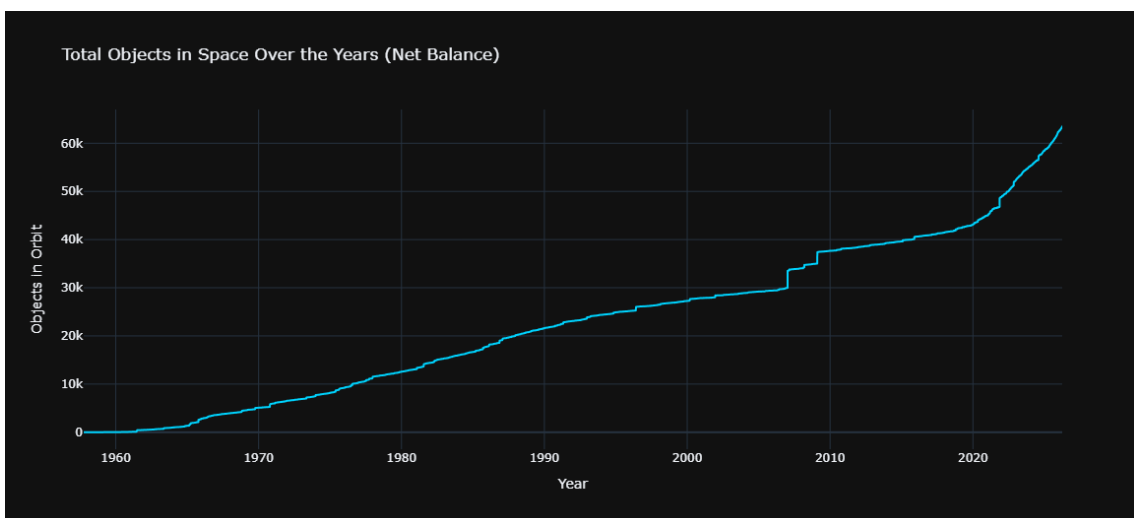


Figure 15: Total Objects in Space Over the Years

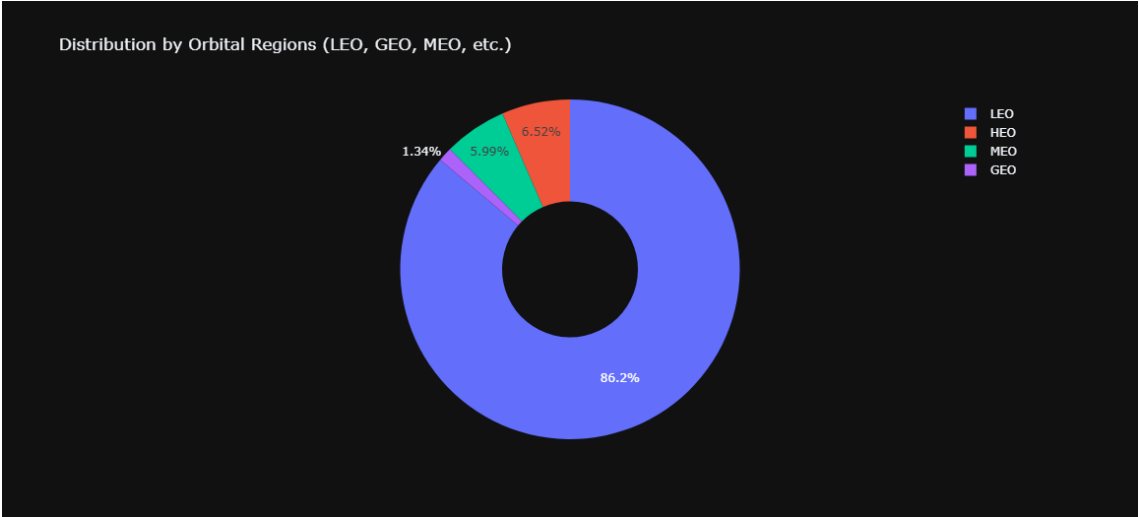


Figure 16: Distribution by Orbital Regions

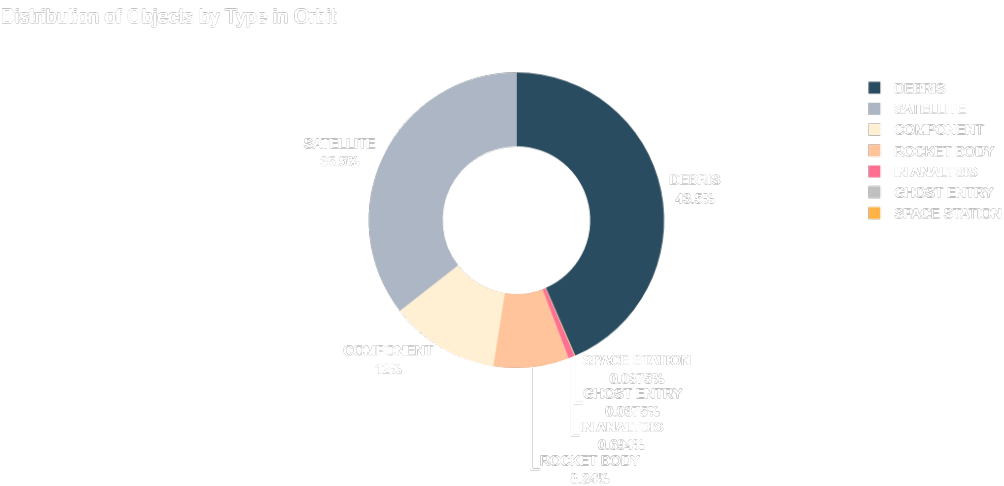


Figure 17: Distribution of Objects by Type in Orbit

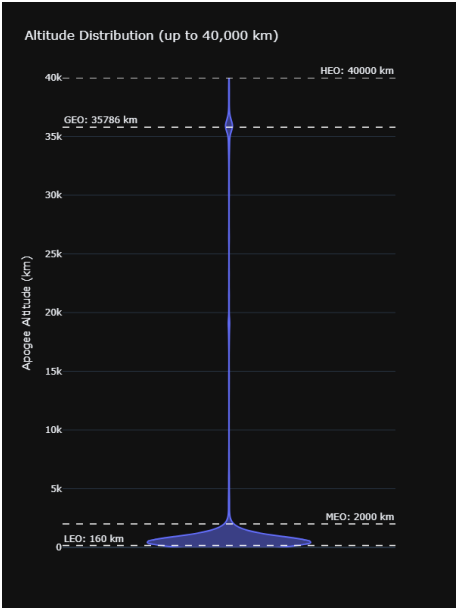


Figure 18: Overall Altitude Distribution

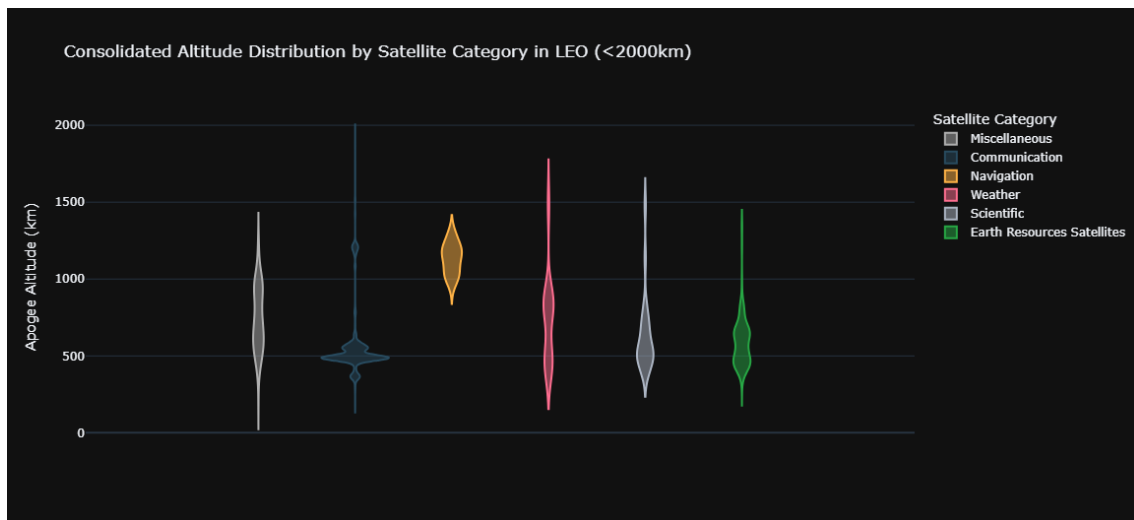


Figure 19: Consolidated Altitude Distribution by Satellite Category in LEO

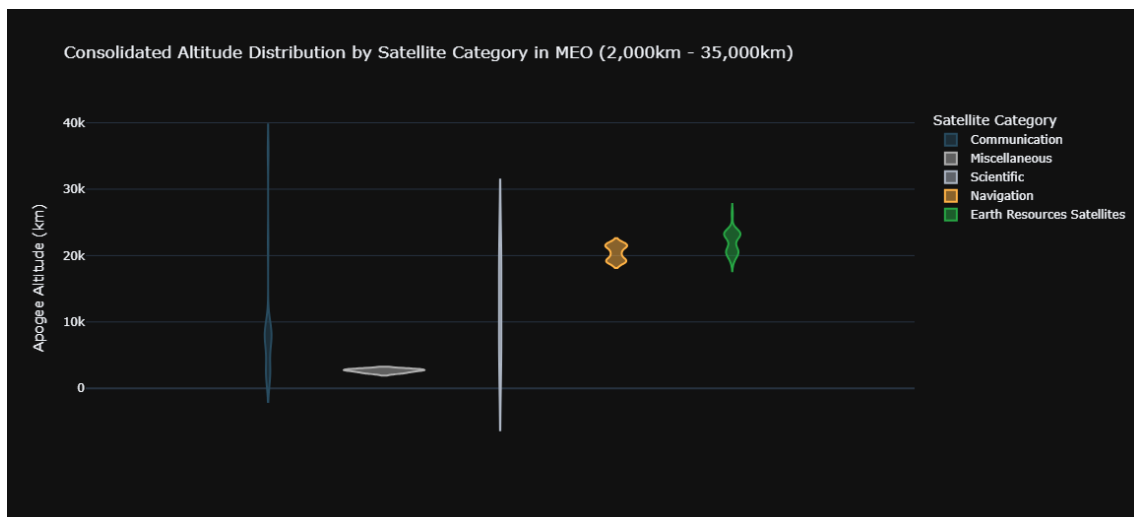


Figure 20: Consolidated Altitude Distribution by Satellite Category in MEO

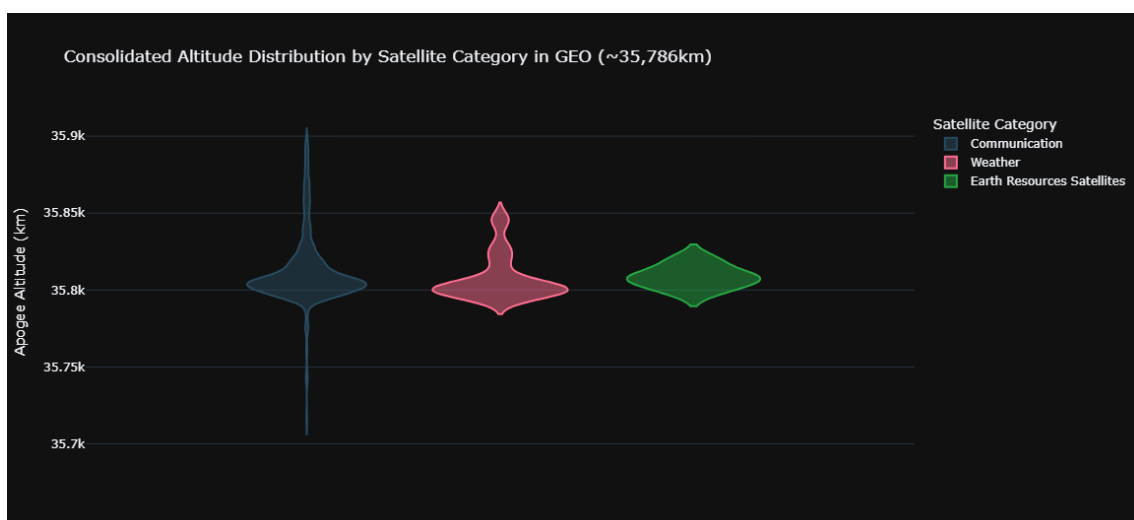


Figure 21: Consolidated Altitude Distribution by Satellite Category in GEO

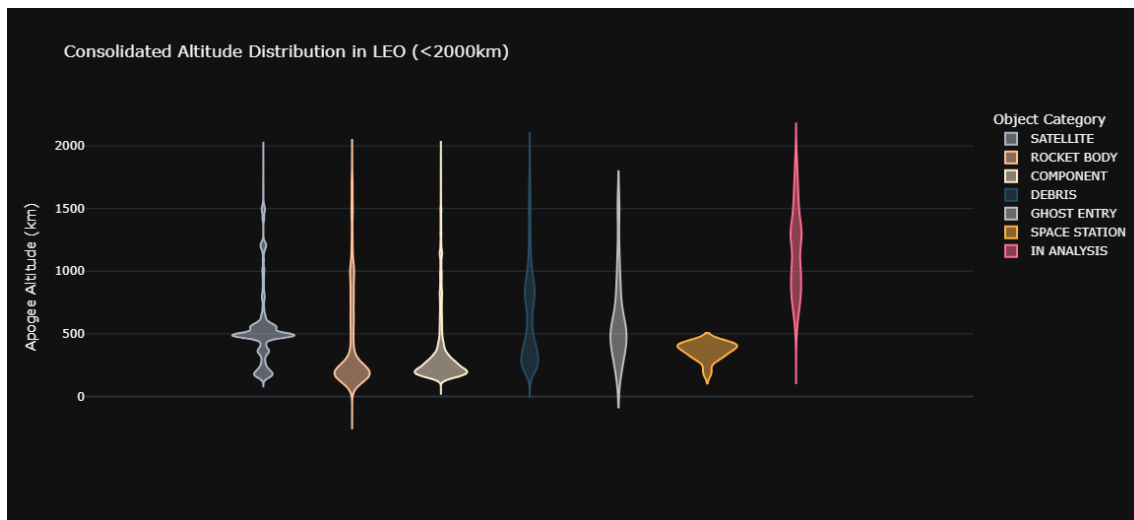


Figure 22: Consolidated Altitude Distribution in LEO

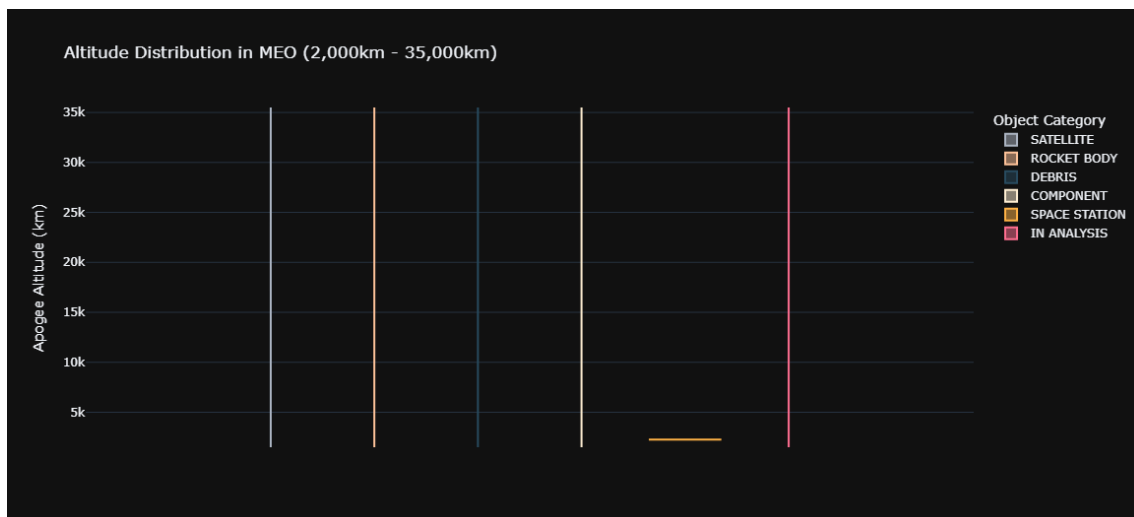


Figure 23: Consolidated Altitude Distribution in MEO

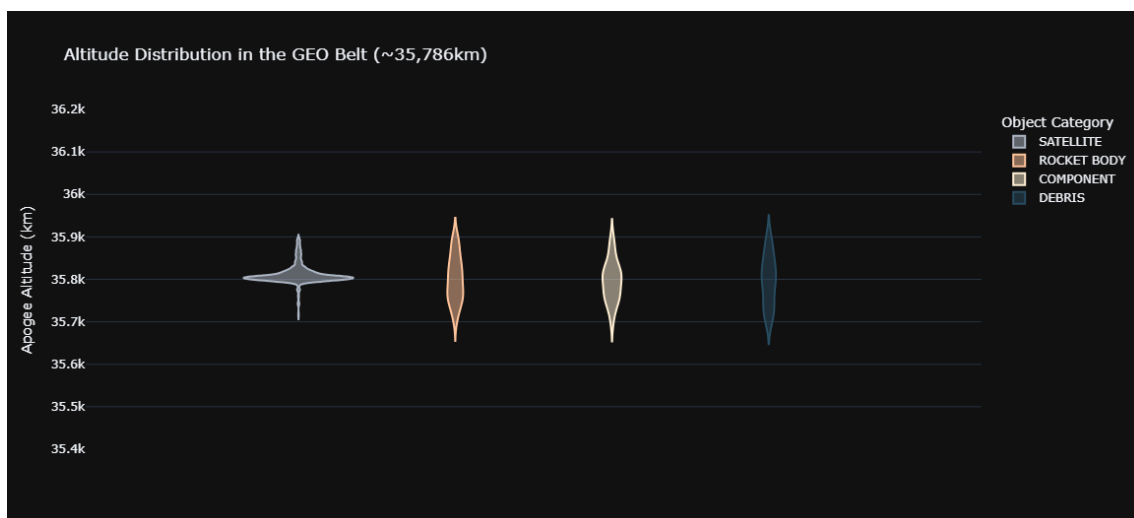


Figure 24: Consolidated Altitude Distribution in GEO

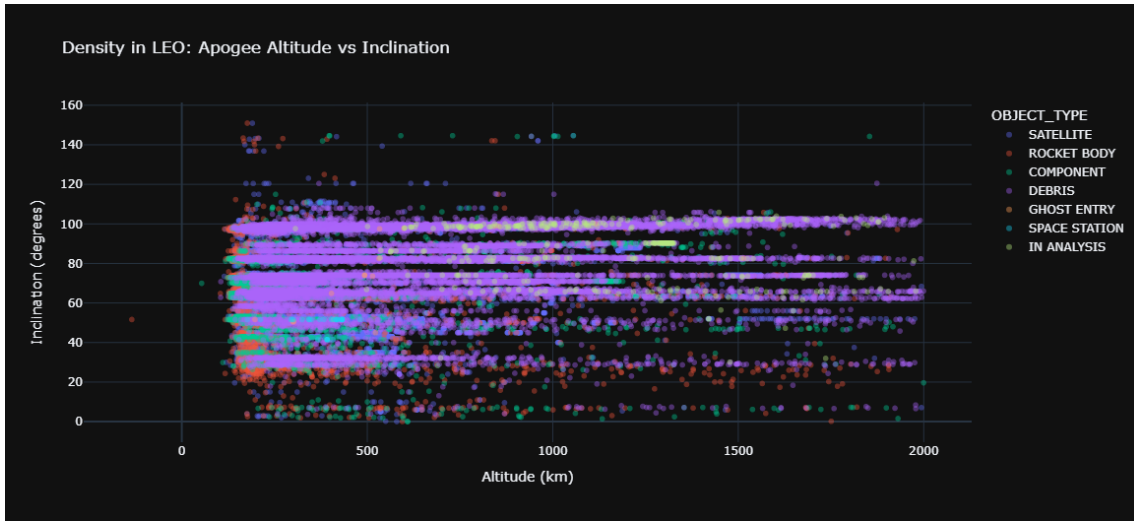


Figure 25: Density in LEO: Apogee Altitude vs Inclination

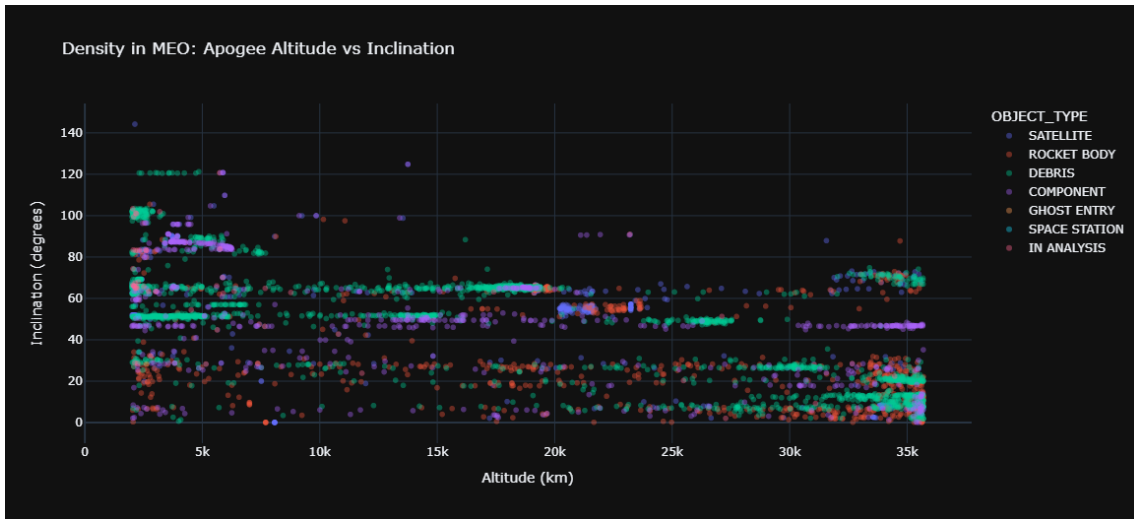


Figure 26: Density in MEO: Apogee Altitude vs Inclination

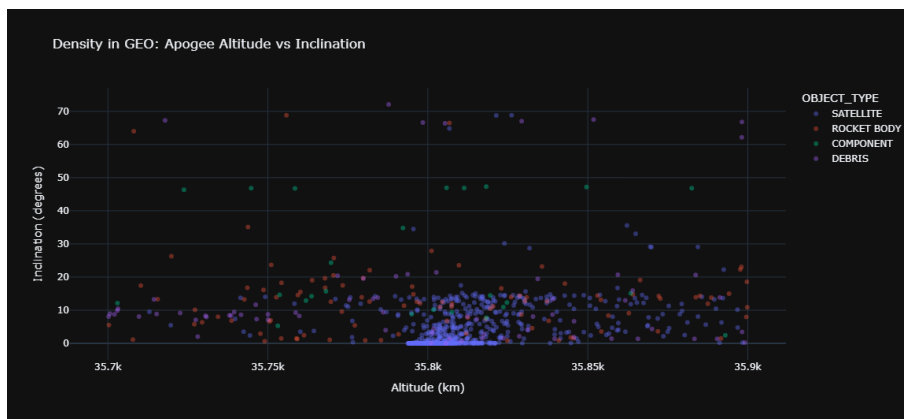


Figure 27: Density in GEO: Apogee Altitude vs Inclination

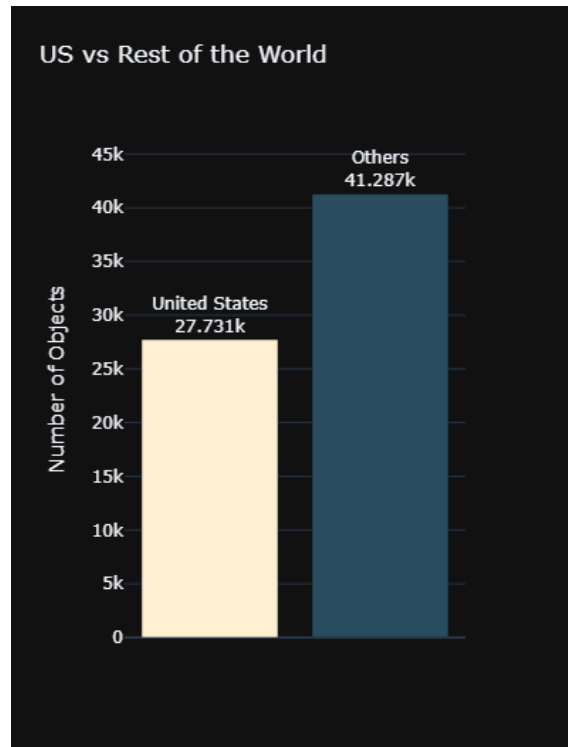


Figure 28: Satellite Ownership: US vs Rest of the World

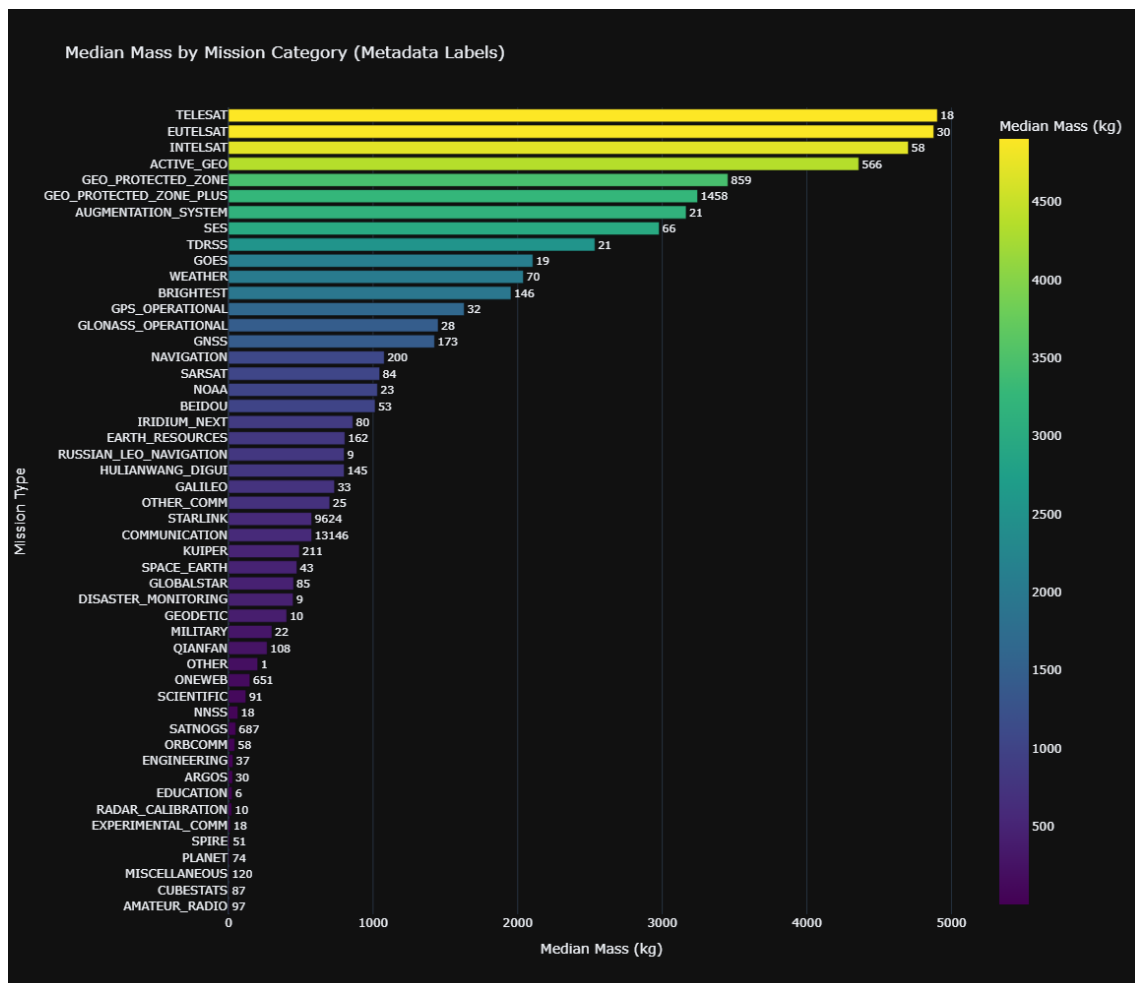


Figure 29: Median Mass by Mission Category (Metadata Labels)

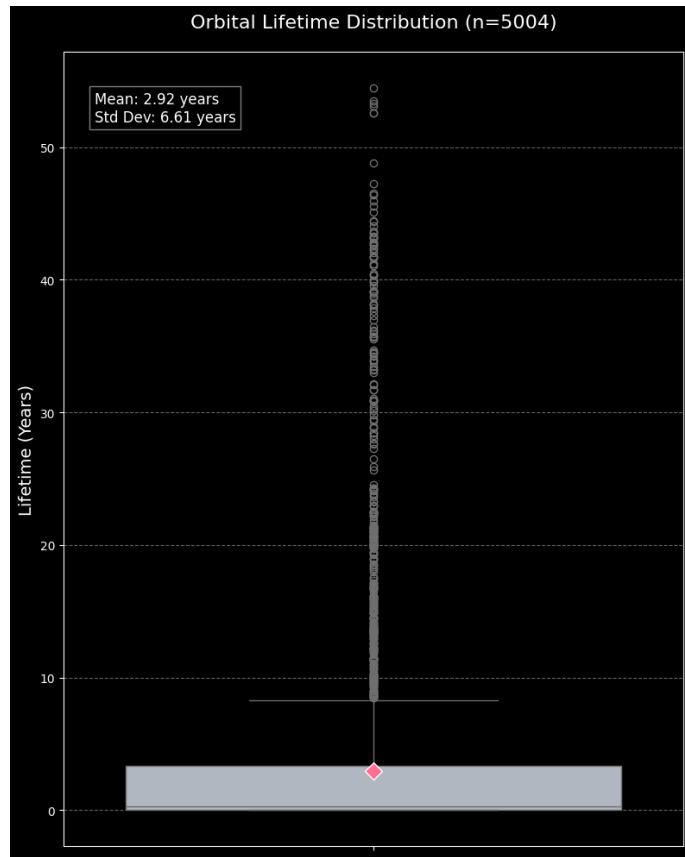


Figure 30: Orbital Lifetime Distribution

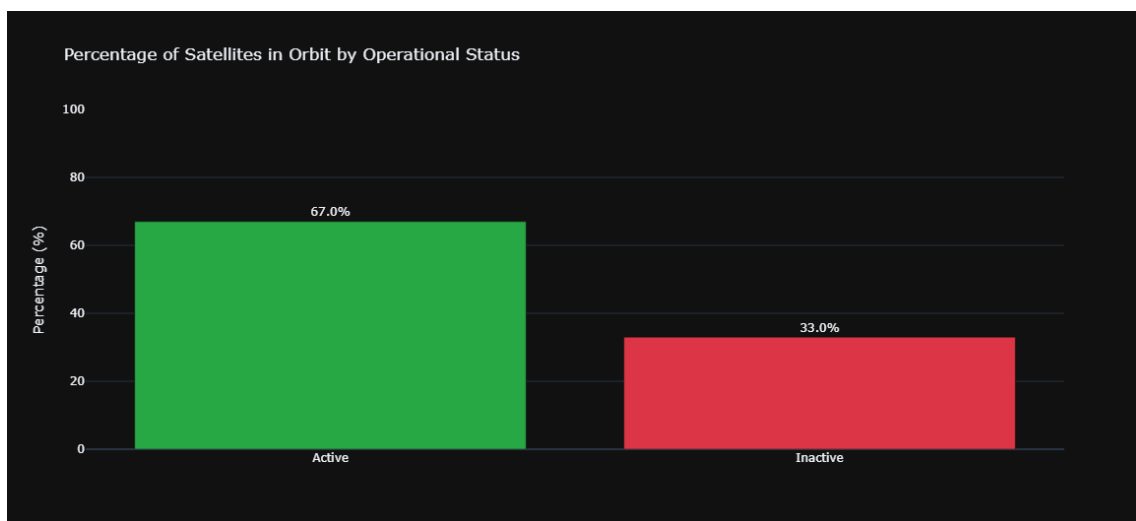


Figure 31: Percentage of Satellites in Orbit by Operational Status

Top 10 Nearest Space Objects to ISS (Next 7 Days)			
NAME	NORAD_ID	MIN_DIST_KM	TIME_UTC
HST	20580	8.7	2026-04-05 03:30:00
STARLINK-2613	48353	18.15	2026-04-11 15:30:00
STARLINK-2647	48658	46.15	2026-04-06 11:15:00
STARLINK-2754	48641	54.24	2026-04-10 05:15:00
STARLINK-2389	47808	60.28	2026-04-07 08:00:00
STARLINK-1281	45370	63.12	2026-04-11 14:15:00
STARLINK-2717	48659	68.12	2026-04-08 17:15:00
STARLINK-2161	47747	68.26	2026-04-10 11:45:00
STARLINK-2232	48578	69.27	2026-04-07 04:30:00
STARLINK-4188	52843	73.85	2026-04-11 02:15:00

Figure 32: Top 10 Nearest Space Objects to ISS (Next 7 Days)

Top Launch Sites Ranking

Launch Site	Total Launches
Cape Canaveral	14002
Plesetsk Cosmodrome	13623
Baikonur Cosmodrome	12744
Vandenberg SFB	12513
Taiyuan Satellite Launch Center	6521
Guiana Space Centre	2130
Jiuquan Satellite Launch Center	1225
Satish Dhawan Space Centre	1158
Xichang Satellite Launch Center	910
Tanegashima Space Center	565
Vostochny Cosmodrome	508
Kennedy Space Center	404

Figure 33: Top Launch Sites Ranking by Total Launches

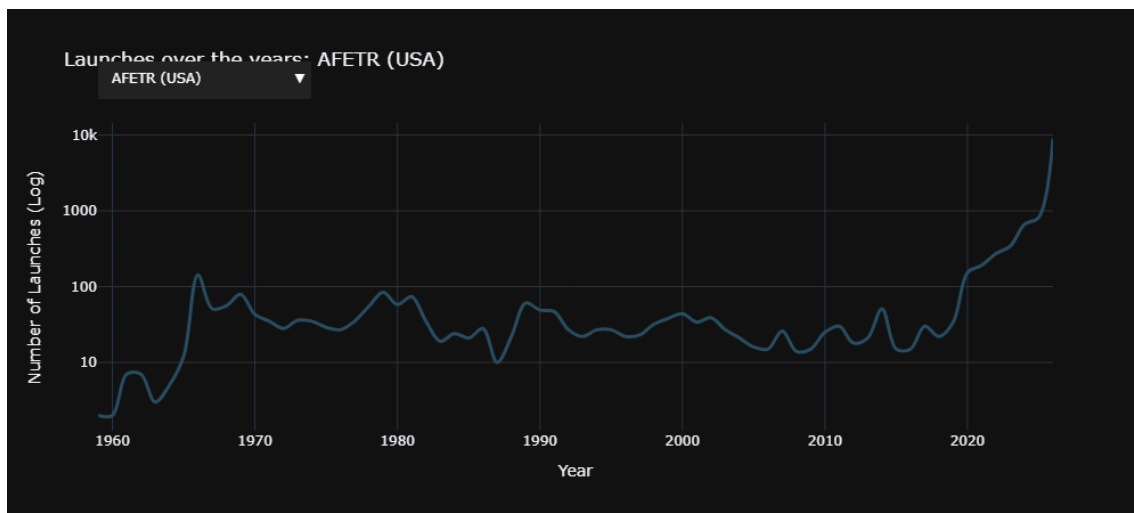


Figure 34: Launches Over the Years: AFETR (USA)

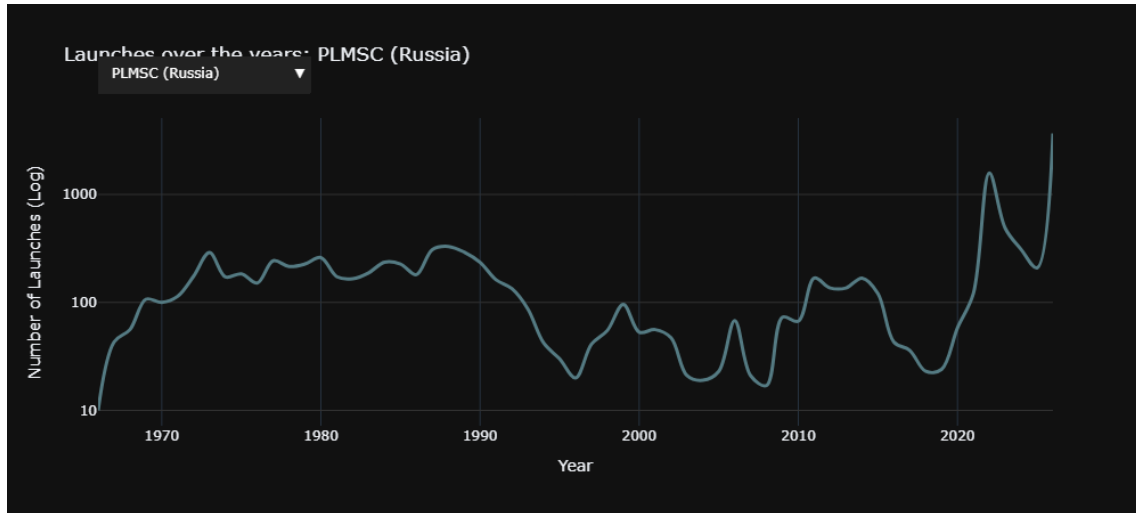


Figure 35: Launches Over the Years: PLMSC (Russia)

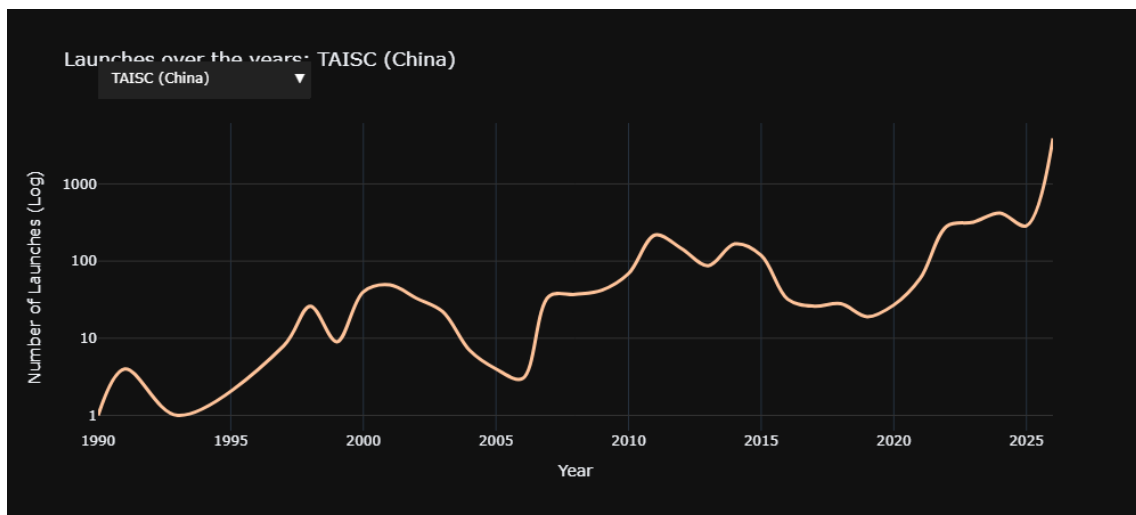


Figure 36: Launches Over the Years: TAISC (China)